

Bingran Li

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SUMMARY

- From a humanities high school student to a mathematics undergraduate interested in the theory and application of optimization, probability and computation, into real problems I am passionate about.
- My guiding principle is to distill the essence of real-world problems, formulate them precisely, and solve them using minimal yet effective mathematics.

EDUCATION

The Chinese University of Hong Kong, Shenzhen

2022 – 2026 (Expected)

Bachelor of Science in Mathematics and Applied Mathematics

- Cumulative GPA: **3.923/4.000** Rank: **1st/79**
- **Main Courses:** Real Analysis(A), Numerical Analysis, Functional Analysis(A), Advanced Linear Algebra, Honours Ordinary Differential Equations(A), Partial Differential Equations(A), Stochastic Processes(A), Mathematical Statistics, Modern Optimization, Optimization in Machine Learning and Data Science
- **Postgraduate Courses:** Topics in Financial Mathematics(A), Measure Theoretic Probability(A-), Numerical Linear Algebra(A), Optimization in Machine Learning(A), Advanced Convex Optimization(A), Analysis of Algorithms, Computer Experiments and Uncertainty Qualification, Advanced Stochastic Processes (Audited)

The Chinese University of Hong Kong

Jun. 2024 – Aug. 2024

International Summer School

WORKING PAPERS AND PUBLICATIONS

Adam Converges Without Any Modification On Update Rules

Yushun Zhang, **Bingran Li**, Ruoyu Sun, Zhi-Quan Luo

To be submitted, 2025

RESEARCH EXPERIENCE

Memory Mechanism in Machine Learning Systems

Sep. 2025 – Present

- Explore the memory mechanism in machine learning systems, such as transformers and modern RNNs

Implementation of an Interior Point Method(IPM)

Apr. 2025 – Jun. 2025

Supervised by Prof. Yin Zhang & Prof. Yurii Nesterov

- Implement the Asymmetric Long-Step Primal-Dual IPMs with Dual Centering on Linear Programming problems

Stochastic Optimization in Deep Learning

Feb. 2025 – Present

Supervised by Prof. Ruoyu Sun & Mr. Yushun Zhang

- Prove that Adam under with-replacement sampling with $\beta_1 > 0$ can converge when β_2 is large enough, which provides guidance for hyperparameter tuning in LLM pre-training

Algorithmic Complexity in Markov Decision Processes

Oct. 2024 – Jan. 2025

Supervised by Prof. Yinyu Ye

- Explore to prove the strongly polynomial time complexity for the policy iteration method for deterministic Markov decision processes

Course Project (DDA6202 Optimization in Machine Learning)

Fall 2024

Title: *Analyzing Non-diagonal Preconditioning Optimizers in Deep Learning*

- Analyze and compare the performance between diagonal preconditioning optimizers (e.g. Adam-mini) and non-diagonal preconditioning optimizers (e.g. SOAP, Shampoo) on linear model, MLP and Transformer
- Try to invent a new optimizer for deep learning. No significant improvement has been observed yet.

Concentration Inequality in Reinforcement Learning

Aug. 2024 – May. 2025

Supervised by Dr. Yingru Li

- Apply a sequential random projection method 1) to approximate the posterior variance of independent Gaussian arm bandits, and 2) to a non-asymptotic probability bound for discrete stochastic integral

Stochastic Control for Portfolio Selection

Oct. 2023 – Jul. 2024

Supervised by Prof. Yang Liu

Title: *Market-based Value-at-Risk Constraints in Hedge Funds*

- Develop a closed-form optimal strategy for Merton’s problem with S-shape utility under a market-based Value-at-Risk constraint

READING GROUPS

Concentration Inequalities
Jan. 2024 – Jul. 2024

Supervised by Prof. Jeff Yao

- Delivered lectures on the Johnson-Lindenstrauss Lemma and [Transportation Method](#)

Graph Theory
Jan. 2025 – May 2025

Supervised by Prof. Kenneth Wing Ki Shum

- Delivered lectures on Network Flow problems and Random Graphs

Analytic Number Theory
Jan. 2025 – Mar. 2025

Supervised by Prof. Tiefeng Jiang

- Delivered lectures on [Arithmetic Function Theory](#)

Theoretical Data Science
Feb. 2025 – May 2025

Textbook: Foundations of Data Science, Supervised by Prof. Zhi-Quan Luo

- Delivered presentations on [high dimensional probability and geometry](#)
- Report on [the maximum number of almost orthogonal unit vectors in \$\mathbb{R}^d\$](#)

TEACHING

Lecturer
Sept. 2023 – Present

Academic Department of Statistics Alliance

- MAT2050 Mathematical Analysis Midterm Review, 2023-2024 Fall term
- MAT2004 Mathematical Statistics Midterm Review, 2023-2024 Spring term

Undergraduate Student Teaching Fellow

School of Science and Engineering Help Room
Spring 2024

MAT2001 Honours Ordinary Differential Equations (Instructor: Prof. Wei-Ming Ni)
Fall 2024

STA4001 Stochastic Processes (Instructor: Prof. Tiefeng Jiang)
Spring 2025

STA4001H Honours Stochastic Processes (Instructor: Prof. Tiefeng Jiang)
Fall 2025

INDUSTRY EXPERIENCE & ACTIVITIES

MY Capital
Shanghai, China

Quantitative Research Intern
Jul. 2025 – Aug. 2025

- Portfolio optimization

Quantitative Trading and Investment Association (QTIA)
CUHK-Shenzhen

Executive Committee Member
Jun. 2025 – Present

- Organize seminars on practical quantitative trading topics for members
- Connect with industry professionals to provide networking opportunities

HONORS

Undergraduate Research Awards
2024, 2025

Academic Performance Scholarship (Class B)
2023-2024

National Scholarship for Encouragement
2023

Academic Performance Scholarship (Class C)
2022-2023

Muse College Scholarship
2022, 2023, 2024

SKILLS

- **Technologies:** PyTorch, CUDA (Certificate), Numba, MOSEK, COPT, Vite.js
- **Programming:** Python, MATLAB, C++, Java
- **Tools:** Git, Linux (SSH, command-line), $\LaTeX$, Markdown
- **Languages:** English (TOFEL 104, GRE Verbel 160) , Mandarin (Native)
- **Hobbies:** Philosophy, Dance (Hip-Hop), Piano, Chorus, Chinese Poetry Sculpting